Ex.No. :03

Capturing and Analyzing the Packet Traces of DNS

Aim

To capture and analyzes the request and response packet traces of DNS protocol using Wireshark

DNS

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

How DNS works?

DNS servers convert URLs and domain names into IP addresses that computers can understand and use. They translate what a user types into a browser into something the machine can use to find a webpage. This process of translation and lookup is called DNS resolution.

The basic process of a DNS resolution follows these steps:

1. The user enters a web address or domain name into a browser.
2. The browser sends a message, called a recursive DNS query, to the network to find out which IP or network address the domain corresponds to.
3. The query goes to a recursive DNS server, which is also called a recursive resolver, and is usually managed by the internet service provider (ISP). If the recursive resolver has the address, it will return the address to the user, and the webpage will load.
4. If the recursive DNS server does not have an answer, it will query a series of other servers in the following order: DNS root name servers, top-level domain (TLD) name

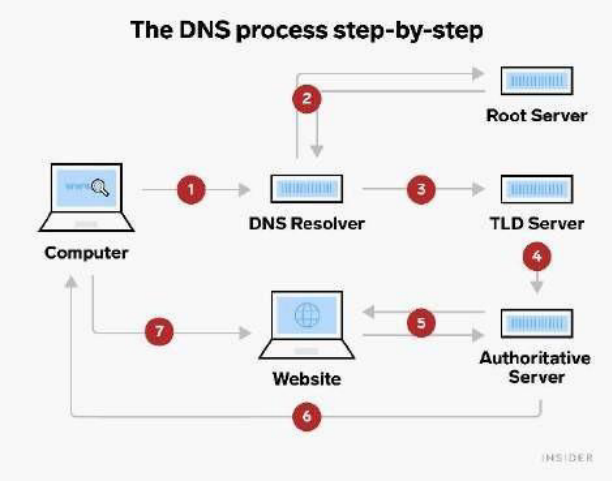
servers and authoritative name servers.

5. The three server types work together and continue redirecting until they retrieve a DNS record that contains the queried IP address. It sends this information to the recursive

DNS server, and the webpage the user is looking for loads. DNS root name servers and TLD servers primarily redirect queries and rarely provide the resolution themselves.

6. The recursive server stores, or caches, the A record for the domain name, which contains the IP address. The next time it receives a request for that domain name, it can respond directly to the user instead of querying other servers.
7. If the query reaches the authoritative server and it cannot find the information, it returns an error message.

The entire process querying the various servers takes a fraction of a second and is usually imperceptible to the user.



DNS structure

- The domain name is usually contained in a URL. A domain name is made of multiple parts, called labels. The domain hierarchy is read from right to left with each section denoting a subdivision.

- The TLD appears after the period in the domain name. Examples of top-level domains include .com, .org and .edu, but there are many others. Some may denote a country code or geographic location, such as .us for the United States or .ca for Canada.

- Each label on the left-hand side of the TLD denotes another subdomain of the domain to the right. For example, in the URL www.techtarget.com, "techtarget" is a subdomain of .com, and "www." is a subdomain of techtarget.com.

- There can be up to 127 levels of subdomains, and each label can have up to 63 characters. The total domain character length can have up to 253 characters. Other rules include not starting or ending labels with hyphens and not having a fully numeric TLD name.

DNS server types

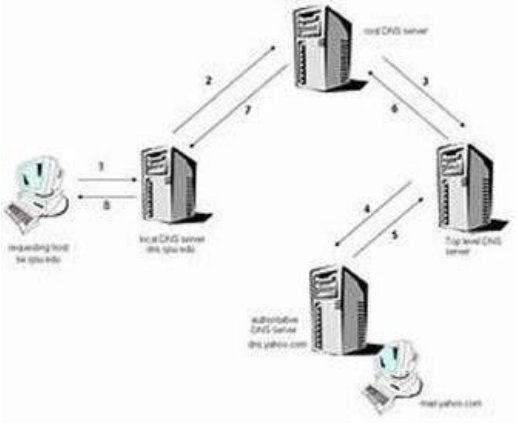
There are several server types involved in completing a DNS resolution. The following list describes the four name servers in the order a query passes through them. They provide the domain name being sought or referrals to other name servers.

1. **Recursive server.** The recursive server takes DNS queries from an application, such as a web browser. It's the first resource the user accesses and either provides the answer to the query if it has it cached or accesses the next-level server if it doesn't. This server may go through several iterations of querying before returning an answer to the client.
2. **Root name server.** This server is the first place the recursive server sends a query if it doesn't have the answer cached. The root name server is an index of all the servers that will have the information being queried. These servers are overseen by the Internet Corporation for Assigned Names and Numbers, specifically a branch of ICANN called the Internet Assigned Numbers Authority.
3. **TLD server.** The root server directs the query based on the top-level domain -- the .com, .edu or .org in the URL. This is a more specific part of the lookup.
4. **Authoritative name server.** The authoritative name server is the final checkpoint for the DNS query. These servers know everything about a given domain and deal with the subdomain part of the domain name. These servers contain DNS resource records with specific information about a domain, such as the A record. They return the necessary record to the recursive server to send back to the client and cache it closer to the client for future lookups.

Types of DNS queries

The following types of DNS queries are the main ones that take place at different points in the DNS resolution:

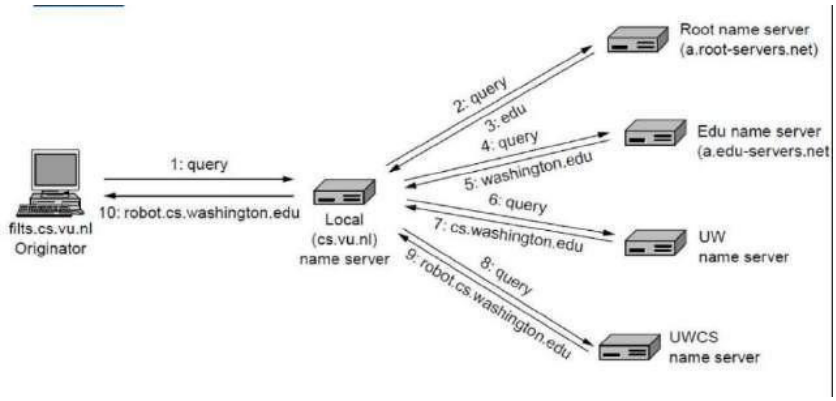
- **Recursive DNS queries** are those that take place between the recursive server and the client. The answer provided is either the full name resolution or an error message saying that the name cannot be found. Recursive queries end in either the answer or an error.



- **Iterative DNS queries** take place between the recursive resolver, which is a local DNS server, and the nonlocal name servers, like the root, TLD and authoritative name servers. Iterative queries do not demand a name resolution; the name servers may instead respond with a referral. The root server refers the recursive server to the TLD, which refers it to an authoritative server. The authoritative server provides the domain name to the recursive server if it has it. Iterative queries resolve in either an answer or a referral.
- **Nonrecursive queries** are those for which the recursive resolver already knows where to get the answer. The answer is either cached on the recursive server or the recursive server knows to skip the root and TLD servers and go directly to a specific authoritative server. If a recursive resolver has cached an IP address from a previous session and serves that address upon the next request, that is considered a nonrecursive query.

Common DNS records

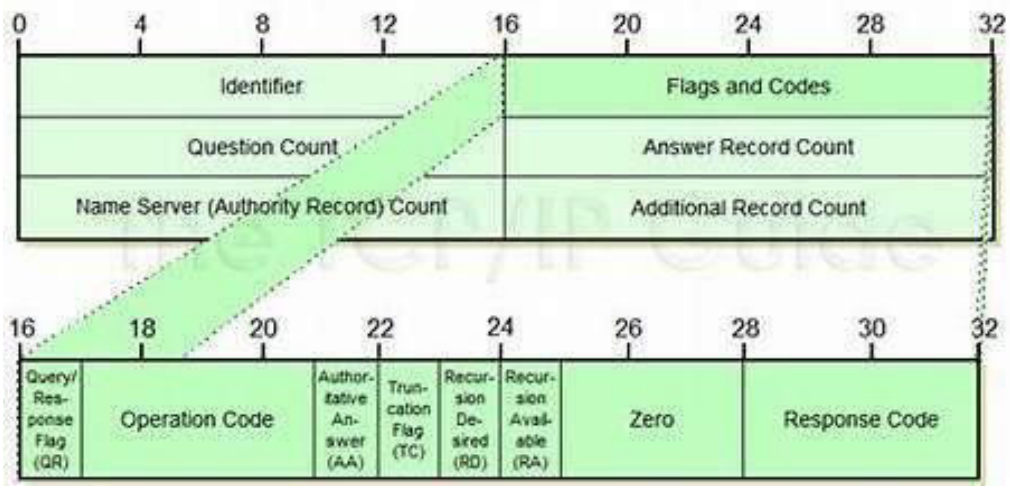
There are many DNS record types, each with their own purpose in denoting how a query should be treated. Common DNS records are the following:

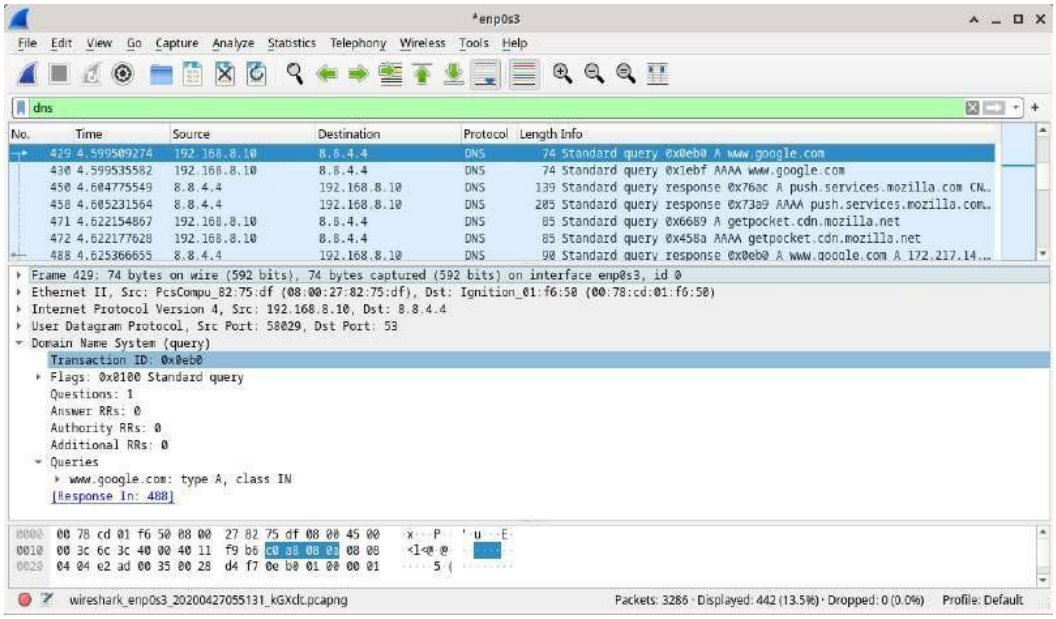


1. **A record.** This stands for address and holds the IP address of a domain. A records only apply to IPv4 addresses. IPv6 addresses have AAAA records instead, which use the longer format of IPv6 addresses. Most websites only have one A record, but some larger sites have several, which helps with load balancing by serving different A records to different users in heavy traffic.
2. **NS record.** These name server records denote which authoritative server is responsible for having all the information about a given domain. Often, domains have both primary and backup name servers to increase reliability, and multiple NS records are used to direct queries to them.
3. **TXT record.** TXT records enable administrators to enter text into DNS. The original purpose was to put human-readable notes in DNS, but today, machine-readable notes are often put there. TXT records are used to confirm domain ownership, secure email and counter email spam.

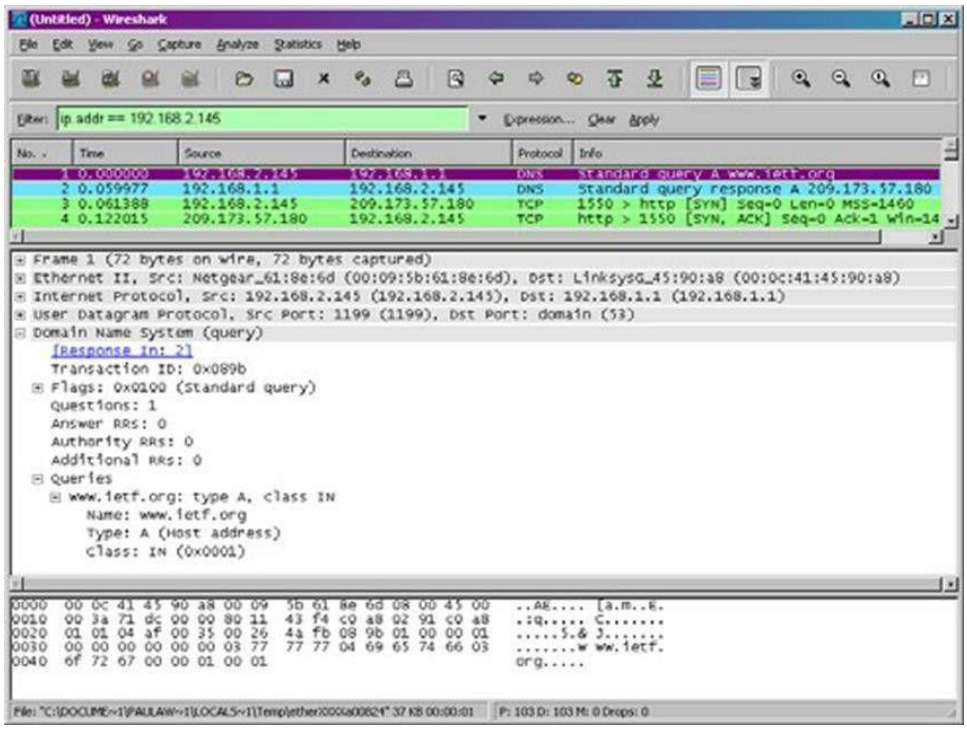
4. **CNAME record.** Canonical name records are used instead of an A record when there is an alias. They are used to retry the query of the same IP address with two different domains. An example would be in the URL searchsecurity.techtarget.com, where the CNAME would query techtarget.com.

DNS message format

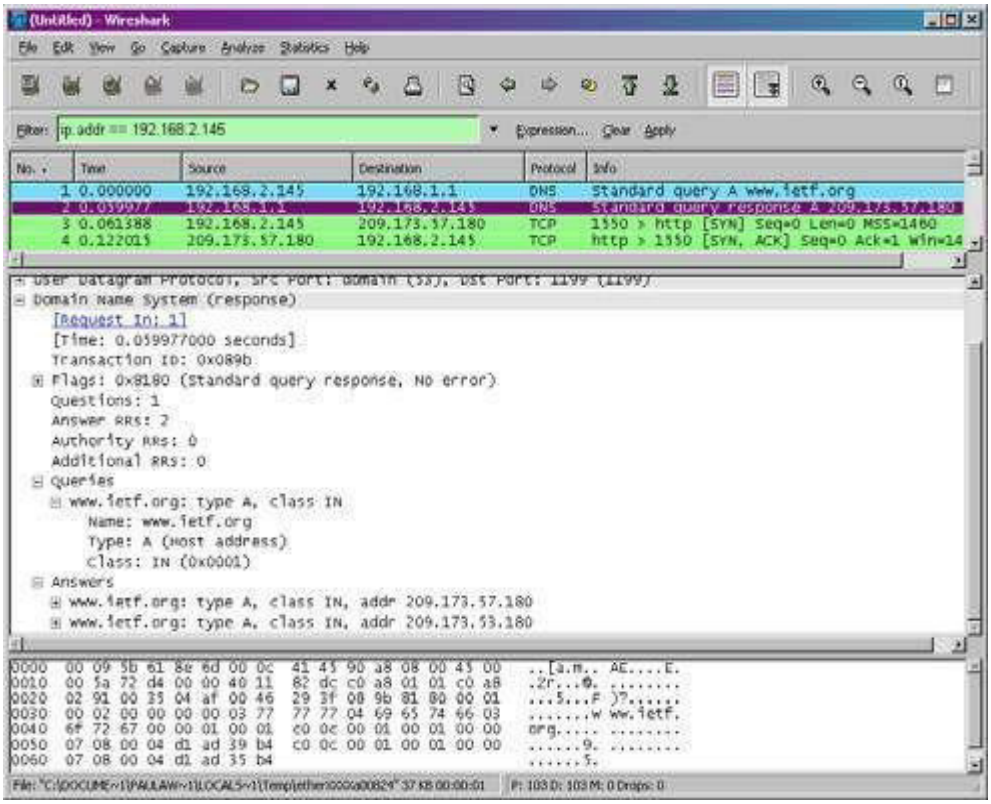




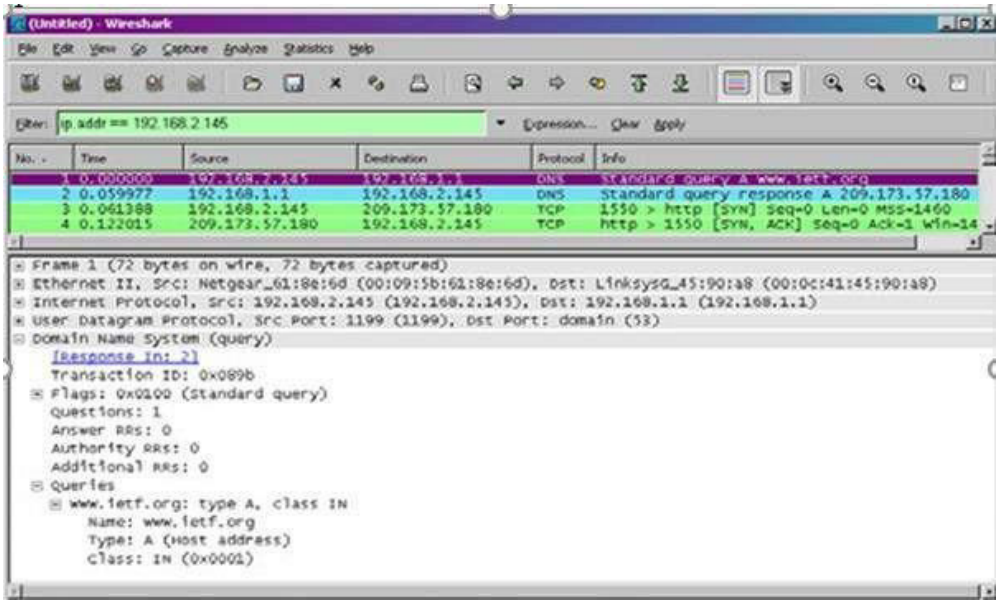
1. Locate the DNS query and response messages. Are then sent over UDP or TCP?
ANSWER :They are sent over UDP



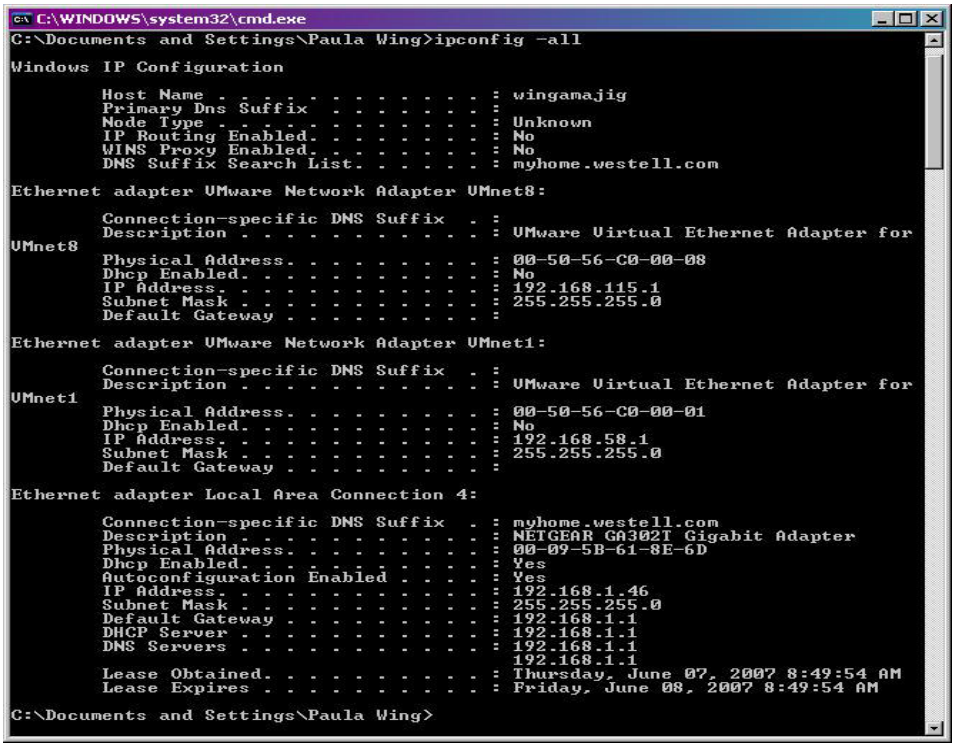
2. What is the destination port for the DNS query message? What is the source port of DNS response message?


(Untitled) - Wireshark
File Edit View Go Capture Analyze Statistics Help
Filter: ip.addr == 192.168.2.145
Expression... Clear Apply
No. Time Source Destination Protocol Info
1 0.000000 192.168.2.145 192.168.1.1 DNS Standard query A www.ietf.org
3 0.061388 192.168.2.145 209.173.57.180 TCP 1550 > http [SYN] Seq=0 Len=0 MSS=1460
4 0.122015 209.173.57.180 192.168.2.145 TCP http > 1550 [SYN, ACK] Seq=0 Ack=1 Win=14
User Datagram Protocol, Src Port: domain (53), Dst Port: 1199 (1199)
Domain Name System (response)
[Request In: 1]
[Time: 0.059977000 seconds]
Transaction ID: 0x089b
Flags: 0x8180 (Standard query response, No error)
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 0
Queries
www.ietf.org: type A, class IN
Name: www.ietf.org
Type: A (Host address)
Class: IN (0x0001)
Answers
www.ietf.org: type A, class IN, addr 209.173.57.180
www.ietf.org: type A, class IN, addr 209.173.53.180
0000 00 09 5b e1 8e 6d 00 0c 41 45 90 a8 08 00 45 00 ..[a.m.. AE....E.
0010 00 5a 72 d4 00 00 40 11 82 dc c0 a8 01 01 c0 a8 .Zr...@. ....
0020 02 91 00 35 04 af 00 46 29 3f 08 9b 81 80 00 01 ...5...F)?......
0030 00 02 00 00 00 00 03 77 77 77 04 69 65 74 66 03 ....w ww.ietf.
0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01 00 00 org..... ....
0050 07 08 00 04 d1 ad 39 b4 c0 0c 00 01 00 01 00 00 ......9. ....
0060 07 08 00 04 d1 ad 35 b4 ......5.

ANSWER: The destination port for the DNS query is 53 and the source port of the DNS response is 53.



3. To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?
ANSWER: It's sent to 192.168.1.1, which is the IP address of one of my local DNS servers.



4. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

ANSWER: It’s a type A Standard Query and it doesn’t contain any answers.

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

ANSWER: There were 2 answers containing information about the name of the host, the type of address, class, the TTL, the data length and the IP address.

Result:

Thus, capturing and analyzing DNS packets using Wireshark has been completed successfully.